

Subject Description Form

Subject Code	EIE1005
Subject Title	Fundamental AI and Data Analytics
Credit Value	2
Level	1
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	Data analytics is for extracting valuable knowledge from raw data. It is very critical to industry, business, and government. Artificial intelligence and machine learning are widely used to automate data analytics processes. This subject aims to provide students with the fundamental concepts of artificial intelligence and data analytics. In particular, it offers hands-on experiences and case studies in the applications of AI and data analytics, with a strong emphasis on their applications in the context of innovation and technological development. It also helps students appreciate how data analytics and artificial intelligence influence our daily life.
Intended Subject Learning Outcomes	Upon completion of the subject, students will be able to: 1. Demonstrate an understanding of the foundational concepts of Artificial Intelligence and Data Analytics (AIDA); 2. Acquire basic skills in using AIDA technologies and applications; 3. Articulate examples of how the adoption AIDA could enhance their chosen disciplines and contribute to technological progress; 4. Demonstrate an awareness of global contemporary ethical issues and impact from AIDA applications in daily life.
Subject Synopsis/ Indicative Syllabus	1. <u>Introduction</u> • Introduction to AI and data analytics • Introduction to machine learning • Ethical issues and impact from AI and data analytics applications 2. <u>Understanding and Presenting Data</u> • Data quality and preprocessing • Data analysis – Basic statistics, clustering, frequent pattern mining • Data visualization 3. <u>AI and Data Analytics in the Virtual World</u> • Development of game AI • Reward scheduling via data analytics • AI and data analytics in the Metaverse • Developing game AI with Unity ML-Agents 4. <u>AI and Data Analytics for Computer Vision</u> • Machine learning for computer vision • Predictive analytics for computer vision • Feature extraction • Pattern recognition

	• Convolutional neural network • Developing simple computer vision systems 5. <u>Conversational AI and Data Analytics</u> • Prescriptive analytics for human-computer interaction • Spoken language technologies • Natural language understanding • Understanding users’ behaviours via data analytics • AI Chatbots																																														
Teaching/Learning Methodology	e-Learning Module: The e-learning module is developed and delivered by the Department of Computing at PolyU, consisting of readings, exercises and assessments that are designed to introduce students to the basic concept and practice of AIDA. The e-learning module will provide basic foundation concepts about AIDA, as well as their potential global and societal context impacts. A brief understanding about the technology and applications will also be provided. Students are required to successfully complete the e-learning module (including video watching, an after-class exercise, and a lab with the AIDA interactive playground) within the first seven weeks of the semester in which they are taking the subject. Besides, online materials and video demonstrations are prepared for each of the software tools used in the workshops. Students can gain the background information about the tools and the skill to use them before they join the Tutorial Workshops. Lectures: The subject matters will be delivered through lectures. Students will be engaged in the lectures through Q&A, discussions, and specially designed classroom activities. Tutorial Workshops: Students will work on three AI and data analytics workshops using software tools. In each workshop, students may start from doing some small and easy tasks at the first stage. Students will solve a simple real-world problem commonly found in daily life in the second stage, using the knowledge and know-how that they have learnt from the first stage and the lectures. Tutorials are conducted in an interactive manner through discussions between students and tutors on the problems encountered during the workshop activities. The subject will offer the datasets used in the workshop. Assignment and Demonstration: Students will write a report for each workshop and demonstrate their work for one of the three workshops.																																														
Assessment Methods in Alignment with Intended Subject Learning Outcomes	<table><tr><th rowspan="2">Specific Assessment Methods/Tasks</th><th rowspan="2">% Weighting</th><th colspan="4">Intended Subject Learning Outcomes to be Assessed (Please tick as appropriate)</th></tr><tr><th>1</th><th>2</th><th>3</th><th>4</th></tr><tr><td>1. Continuous Assessment (total: 100%)</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>• Workshop reports</td><td>40%</td><td>✓</td><td>✓</td><td>✓</td><td></td></tr><tr><td>• Workshop demonstration</td><td>15%</td><td>✓</td><td>✓</td><td></td><td></td></tr><tr><td>• e-learning module and seminars</td><td>10%</td><td>✓</td><td>✓</td><td></td><td>✓</td></tr><tr><td>• Test and quizzes</td><td>35%</td><td>✓</td><td>✓</td><td></td><td>✓</td></tr><tr><td>Total</td><td>100%</td><td></td><td></td><td></td><td></td></tr></table>	Specific Assessment Methods/Tasks	% Weighting	Intended Subject Learning Outcomes to be Assessed (Please tick as appropriate)				1	2	3	4	1. Continuous Assessment (total: 100%)						• Workshop reports	40%	✓	✓	✓		• Workshop demonstration	15%	✓	✓			• e-learning module and seminars	10%	✓	✓		✓	• Test and quizzes	35%	✓	✓		✓	Total	100%				
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	Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Tutorial Workshops will require students to apply what they have learnt to solve problems. There will be open-ended tasks that allow students to exercise their creativity in solution design. In the report, students need to write a reflection on how AIDA can be applied to their disciplines. Three workshop reports weight 40% of the whole assessment. In the workshop demonstration, students will think critically to discuss various aspects in the demonstration. The demonstration will be assessed individually and weights 15% of the whole assessment. Each group member will demonstrate the part he/she is responsible for in the workshop. Tests will assess students' achievement of the learning outcomes in a more formal manner. There will be a quiz after each workshop and a final test for the whole subject. Tests are assessed individually and weights 35% of the whole assessment. The workshops will use publicly available or proprietary software tools so that programming knowledge and mathematical skills are kept to a minimum, i.e., no programming and high-level mathematical background is assumed. Students will use available data sources, machine learning models, analytic algorithms, and evaluation methods from the tools to solve data analytics problems. Enthusiastic students could use the cloud-based API to perform more complex tasks. E-learning module aims to equip students with the basic concept and practice of AIDA.	
Student Study Effort Expected	Class contact (time-tabled):	
	• Lectures	11 Hours
	• Tutorial Workshops	14 Hours
	• Tests	1 Hour
	Other student study effort:	
	• Lecture: preview/review of notes; e-Learning; Self-studying	24 Hours
	• Workshop assignment and demonstration; preparation for tests	20 Hours
	Total student study effort:	70 Hours

Reading List and References	Reference Materials: 1. J. Moreira, T. Horvath, and A. Carvalho, A General Introduction to Data Analytics, Newark: John Wiley & Sons, Incorporated, 2018. 2. K. Moreland. The ParaView Tutorial, Version 5.6. Technical Report SAND 2018-11803 TR, Sandia National Laboratories, 2018. 3. A. Majumder, <i>Deep Reinforcement Learning in Unity: With Unity ML Toolkit</i>. Berkeley, CA: Apress, 2020. 4. D. Buckley, “Unity ML-Agents Tutorials – Complete Guide,” <i>GameDev Academy</i>, 2022 (online resource). 5. Deconstructing Chatbots: https://www.youtube.com/hashtag/deconstructingchatbots 6. A.R. Freed, <i>Conversational AI: Chatbots that work</i>. Manning, 2021 7. M. McTear, <i>Conversational Ai: Dialogue Systems, Conversational Agents, and Chatbots (Synthesis Lectures on Human Language Technologies)</i>. Morgan & Claypool, 2020 8. E. Ruane, A. Birhane, and A. Ventresque, “Conversational AI: Social and Ethical Considerations”, in <i>Proc. AICS</i>, 2019. 9. J. Lau,B. Zimmerman, and F. Schaub, “Alexa, are you listening?: Privacy perceptions, concerns and privacy-seeking behaviors with smart speakers,” in <i>Proc. Human-Computer Interaction</i>, 2018, pp. 1-31 10. A. Dertat, “Applied Deep Learning – Part 4: Convolutional Neural Networks,” https://towardsdatascience.com/applied-deep-learning-part-4-convolutional-neural-networks-584bc134c1e2 (online resources) 11. Vaibhav Verdhan, <i>Computer Vision Using Deep Learning: Neural Network Architectures with Python and Keras</i>. Apress, 2021.
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